



TASK-04

Coffee Shop Sales Analysis

Submitted to: DEN(Data Science)

Submitted by: Aleem Shoukat(STU-DS-251-689)

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Coffee Shop Sales Analysis

1. Project Overview

This project aimed to **analyze coffee shop sales data** to identify profitable products, detect loss-making items, predict future sales, and develop strategies to improve profit margins. The dataset contained **transaction details** such as date, time, store location, product category, quantity, and prices.

2. Objectives

1. Clean and prepare the dataset for analysis.
 2. Identify top-selling, most profitable, and loss-making products.
 3. Use predictive modeling to estimate profit based on key features.
 4. Perform time series analysis to forecast monthly sales.
 5. Provide data-driven recommendations to increase profit and reduce losses.
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3. Methodology

3.1 Data Cleaning & Preparation

- Converted dates into proper **datetime** format for analysis.
- Created a **Total Price** column ($\text{unit_price} \times \text{transaction_qty}$).
- Assumed cost price as **70% of selling price** to calculate profit.
- Removed duplicates and ensured there were no missing values.
- Saved a cleaned version of the dataset for further use.

3.2 Profitability Analysis

- Calculated **total sales revenue** and **total profit** for each product and category.
- Identified **Top 10 most profitable products** and **Bottom 10 loss-making products**.
- Computed **profit margin (%)** for each category.

3.3 Predictive Modeling (Linear Regression)

- Selected key features:
 - Numeric: Transaction Quantity, Unit Price
 - Categorical: Store Location, Product Category
- Applied **One-Hot Encoding** for categorical features.
- Trained a **Linear Regression model** to predict transaction profit.
- Evaluated performance using **MAE, RMSE, and R² score**.

- Checked **feature importance** to identify main profit drivers.

3.4 Time Series Analysis (ARIMA)

- Aggregated sales into **monthly totals**.
 - Built an **ARIMA (1,1,1)** model to forecast the next 6 months' sales.
 - Plotted historical vs. forecasted sales for visualization.
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4. Key Findings

Top Profitable Products

- Products like *Gourmet Brewed Coffee – Ethiopia Rg* and *Hot Chocolate – Dark Chocolate Lg* had the highest profit margins.

Loss-Making Products

- Certain tea and low-priced coffee items consistently showed **negative profit margins**.

Category Insights

- **Coffee** contributed the highest revenue and profit.
- **Tea** and **Drinking Chocolate** had lower margins compared to Coffee.

Sales Forecast

- Forecast indicates **steady growth in sales** over the next 6 months, with peak sales expected in the holiday season.
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5. Recommendations

To Increase Profit Margins:

1. Focus marketing efforts on **top 10 profitable products**.
2. Introduce **bundled offers** for high-profit items.
3. Review pricing strategy for mid-performing products.

To Reduce Losses:

4. Adjust prices or portion sizes for loss-making products.
5. Reduce inventory and promotions for consistently low-demand items.
6. Optimize staffing and stock based on **sales forecasts**.

6. Conclusion

The analysis revealed **clear opportunities** to improve profitability by promoting top products, adjusting pricing, and using forecasts to optimize operations. Implementing these recommendations will help the coffee shop **increase revenue, reduce losses, and prepare better for future demand**.

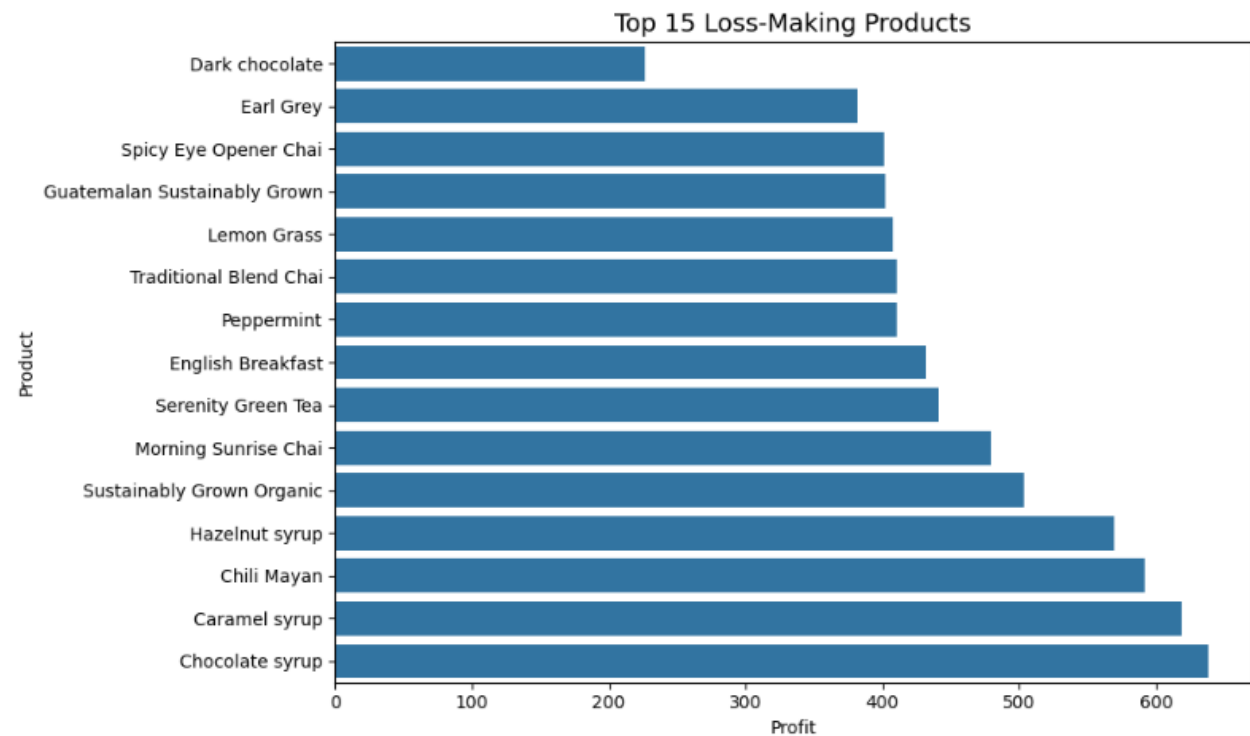
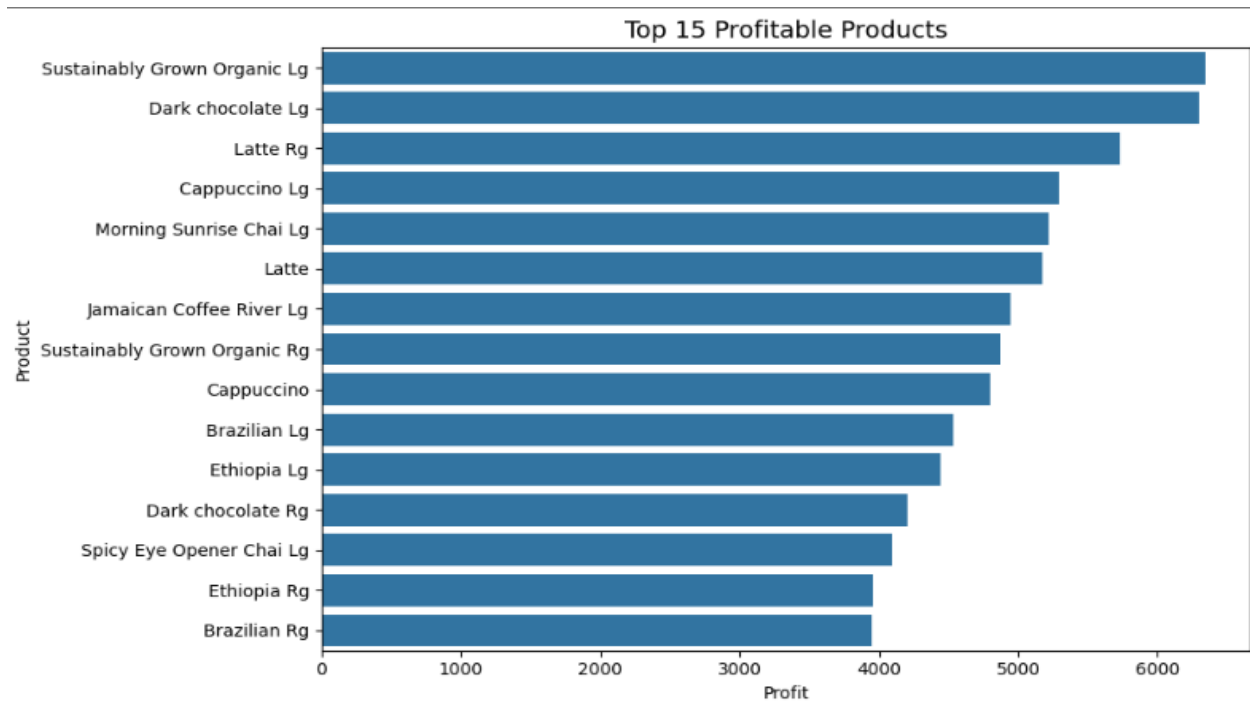
Deliverables Submitted:

- Cleaned dataset (Coffee_shop_sales_cleaned_with_profit.csv)
- Profitability analysis of all products and categories
- Predictive model for profit estimation
- Time series forecast of monthly sales
- Visualizations for top-selling, profitable, and loss-making products
- Recommendations and strategies report

Coffee Shop Sales – Key Performance Indicators (KPIs)

KPI Name	Value	Description
Total Sales Revenue	\$789,812.08	Total revenue earned from all transactions (Jan–Jun 2023).
Total Profit	\$236,549.63	Net profit after deducting estimated cost price (70% of selling price).
Average Profit Margin	30.0%	Percentage of revenue that turned into profit.
Top-Selling Product	Sustainably Grown Organic Lg	Highest total sales value (\$21,151.75).
Most Profitable Product	Sustainably Grown Organic Lg	Product generating the most profit (\$6,345.53).
Loss-Making Product	Dark chocolate	Product with the lowest profit (\$226.56).
Next Month Sales Forecast (July 2023)	\$182,539.18	Projected sales for July 2023 based on ARIMA model.

Product Performance



Trends & Forecast

